Additional file 1: Pre-registration

Blood metal and nutritional biomarkers did not improve identification of low cognitive performance beyond demographics in older US adults: a pre-registered analysis of NHANES 2011-2014

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Written and dated 2026-08-31, at Gate 2, before any modeling.

Integrity statement

No outcome-predictor relationship has been examined at the time of writing.

Work completed so far covers file availability, variable verification, merging, reserved codes, skip patterns, attrition, outcome construction, internal consistency, and missingness. The only analysis touching the outcome has been its own distribution, its internal consistency, and a comparison of included against excluded participants for selection bias. **No association or discrimination between any biomarker and the outcome has been computed, viewed, or estimated.** The pre-specification claim below is therefore intact.

If any decision in this document is revised after modeling begins, the revision must be recorded here with its date and reason, and the affected result reported as exploratory.

1. Question and hypotheses

Primary question. Among US adults aged 60+, how much predictive value do blood metals and nutritional biomarkers add, beyond demographics, for identifying low cognitive performance?

	Statement
H1, primary	M3 achieves higher cross-validated AUC than M0. Operationally, the bootstrap 95% CI for ΔAUC(M3 $-$ M0) excludes zero.
H0, primary null	ΔAUC(M3 $-$ M0) is not distinguishable from zero.
H2, secondary	Nonlinear learners outperform penalized logistic regression.
H3, secondary	The metals and nutritional blocks contribute non-redundantly: ΔAUC(M3 $-$ M0) exceeds either block alone.
H4, exploratory	Predictive contribution differs across the four cognitive domains.

A null result on H1 is a publishable finding and will be reported as the headline if it occurs.

2. Outcome

Composite of four z-scored tests, averaged where at least 3 of 4 are present, with the lowest quartile taken as the label.

Component	Variable
CERAD immediate recall, 0 to 30	`CFDCST1 + CFDCST2 + CFDCST3`, all three trials required
CERAD delayed recall	`CFDCSR`
Animal fluency	`CFDAST`
Digit symbol substitution	`CFDDS`

Terminology is low cognitive performance. NHANES contains no clinical diagnosis, so the term cognitive impairment is not used anywhere in this project.

Eligibility rule. The pre-specified "$\geq$ 3 of 4 scores" rule is retained. The alternative "$\geq$ 2 of 3 instruments" rule, which reflects that `CFASTAT` counts CERAD as a single administration, classifies only 13 of 3,124 participants differently and is reported as a footnote rather than adopted.

Declared property of the label. The label is defined against pooled-sample quantiles, so a held-out participant's label depends on the full sample distribution. This is definitional rather than predictive leakage: the outcome is explicitly relative, and defining it per-fold would make the target itself move between folds. Stated here so it is visibly a deliberate choice.

3. Analytic sample

Option A, decided 2026-08-31 by both authors. All models are fitted and evaluated on one locked row set: participants with every Block 1 to 3 feature and every covariate observed.

n	**1,784**
Events, low cognitive performance	**431**
Prevalence	0.2416
Events per feature, M3	17.2
Cycle composition	1,123 from 2011-2012, 661 from 2013-2014

The row set is frozen in `data/interim/02_analytic_seqn.csv`. Every model, every sensitivity analysis and every bootstrap uses these rows, so that ΔAUC reflects model content alone and never sample composition.

Rationale. Nothing is imputed, so the primary exposure is never fabricated. The dominant exclusion is the `PBCD_H` one-half subsample, which is a random draw and therefore missing-completely-at-random by design.

Known selection. Excluded participants are slightly older and slightly worse off on every measured axis, all differences under 3 percentage points. This is reported in the limitations. Multiple imputation on the full labelled sample ($n = 3{,}124$) is retained as a pre-specified sensitivity analysis and is the formal check on this.

4. Nested models

Primary nested comparison uses Blocks 1 to 3 only.

Model	Features	*n* features
M0	Demographics	5
M1	Demographics + metals	10
M2	Demographics + nutritional	20
M3	Demographics + metals + nutritional	25

Block 1, demographics. `RIDAGEYR`, `RIAGENDR`, `RIDRETH3`, `DMDEDUC2`, `INDFMPIR`.

Block 2, metals, log-transformed. Lead, cadmium, total mercury, selenium, manganese.

Block 3, nutritional and metabolic. B12, methylmalonic acid, serum folate, RBC folate, vitamin D, HbA1c, HDL, total cholesterol, triglycerides, albumin, serum iron, eGFR, NLR, PLR, SII. B12, MMA and SII enter log-transformed.

Covariates. `BMXBMI`, smoking, drinks per day, PHQ-9, stroke, diabetes, hypertension, high cholesterol. These are **not** features in the primary nested models, because the primary question is stated against demographics.

Pre-specified secondary comparison, M0+. Demographics plus covariates, 13 features, with ΔAUC(M3 $-$ M0+) reported alongside the primary. This is the more conservative benchmark: it asks whether biomarkers add anything beyond what a clinician could record without a laboratory. Declared here in advance precisely so it cannot later look like a specification search.

5. Algorithms and evaluation

Decision	Value
Primary model class	L2-penalized logistic regression
Secondary	Random forest, XGBoost
Dropped in advance	MLP and KNN, at this *n* they are not competitive and would only pad the table
CV scheme	Stratified 10-fold, repeated 5 times
Primary metric	**ΔAUC(M3 $-$ M0), bootstrap 95% CI**
Meaningful-difference threshold	ΔAUC ≥ 0.02, declared in advance
Other metrics	AUC, AUPRC, Brier score, calibration slope and intercept, calibration curve, decision-curve net benefit
Never used for selection	Accuracy. At 24% prevalence, predicting everyone normal scores 76%
Master seed	20260831

Mixture-method benchmark: quantile g-computation, reimplemented in Python and validated once against

R `qgcomp` outside the CV loop. BKMR is excluded, on convergence grounds for binary outcomes and on cost. WQS is deferred and will be added only if the biomarker blocks show signal.

6. Leakage controls

Data leakage is the failure mode that would silently invalidate every number in this paper. Controls, all pre-specified:

1. Every preprocessing step (imputation, scaling, encoding, feature selection) is fitted **inside** each training fold, using sklearn `Pipeline` objects constructed inside the CV loop.
2. `.fit()` and `.fit_transform()` are never called on the full dataset.
3. An automated test asserts that no transformer is fitted outside a fold. It fails the run rather than warning, since under the two-gate structure no human reviews between phases.
4. Row-wise derivations only (log transforms, ratios, eGFR, PHQ-9 sum) were done in Phase 1. These involve no fitted parameters and therefore cannot leak.
5. After the modeling code is written, it is explicitly audited for leakage and the audit reported.

7. Sensitivity analyses, all pre-specified

1. Outcome cutoff at the 10th and 20th percentiles.
2. Continuous composite as a regression outcome.
3. DSST "unable to complete" recoded as low performance. 52 participants in the labelled sample: 27 refused, 18 physical limitation, 6 communication problem, 1 other.
4. English-only test administration. Note this excludes both Spanish (322) and Asian-language (69) administrations, not Spanish alone.
5. Complete-case versus multiple imputation on the full labelled sample.
6. Leave-one-block-out check if ΔAUC is large, to confirm the result is not driven by albumin or eGFR acting as proxies for general illness.

8. Inference versus prediction

Survey-weighted logistic regression using `WTMEC2YR` (halved for two pooled cycles), `SDMVSTRA` and `SDMVPSU` is used for **association** claims and to reproduce published estimates as a validity check. Unweighted machine learning is used for **prediction** claims. Every reported result states which it is. Reproducing a published association is a validity check, not a competing result.

This is cross-sectional data. No causal language appears anywhere: not "leads to", "causes", "improves", "protects against", or "reduces risk of". Associations and predictions only.

9. Environment

Pinned in `requirements.txt`. Python 3.12.2, pandas 3.0.3, numpy 2.2.6, scikit-learn 1.8.0, scipy 1.13.0, pyarrow 25.0.1.

10. Sign-off

Role	Name	Date
Data and modeling lead	Neil Patel	2026-08-31
Clinical and framing lead	Shreyan Kancharla	2026-09-05

Decisions in sections 3 and 4 were taken jointly on 2026-08-31. The M0+ secondary comparison in section 4 is a specification choice by the modeling lead, countersigned by the clinical and framing lead on 2026-09-05.

Amendment 1: revision analyses (2026-11-04, post hoc)

Written and dated 2026-11-04, after peer review, before any of the analyses below were run. Every analysis in this amendment is post hoc. It was requested by the editor or reviewers of *Diagnostic and Prognostic Research* (submission `6b0f095f-9022-4344-ae4f-122a3ff71428`), and every result it produces is reported as exploratory. Nothing in sections 1 to 9 is changed: the primary hypothesis, metric, sample, models and sensitivity analyses stand as written.

What the original pre-registration did and did not specify. H1 was a superiority test (the 95% CI for delta-AUC(M3 - M0) excludes zero). The 0.02 threshold in section 5 was declared in advance but no equivalence or non-superiority test was specified, and no external justification for 0.02 was recorded. The manuscript's description of the primary result as a "precise null" therefore went beyond what was pre-registered, and is withdrawn.

A1. Margin analysis (reviewer 1, comment 1). For delta-AUC(M3 - M0) in the primary analysis and in each pre-specified sensitivity analysis, report the paired-bootstrap 90% CI alongside the 95% CI, using the same 2,000 resamples. For margins delta in 0.01, 0.015, 0.02, report (i) a one-sided non-superiority decision (upper 90% bound below delta, a one-sided test at alpha 0.05 of H0: delta-AUC >= delta), and (ii) a two-sided equivalence decision by two one-sided tests (90% CI inside (-delta, delta)). Also report the smallest margin each analysis excludes, which is its upper 90% bound. No margin is chosen as the answer; all are reported.

A2. M4, biomarkers added to the clinical baseline (reviewer 1, comment 3). M4 = M0+ plus both biomarker blocks (13 + 20 = 33 features). Fitted with the same three algorithms, the same nested CV, seed and locked row set as M0 to M0+. The nested contrast is delta-AUC(M4 - M0+).

A3. All pairwise contrasts (reviewer 1, comment 2). delta-AUC with 95% and 90% paired-bootstrap CIs for every pair among M0, M1, M2, M3, M0+ and M4, under each algorithm, plus each nonlinear learner against logistic regression within every specification. The 0.02 margin is applied symmetrically: a difference in either direction whose CI does not extend beyond 0.02 in magnitude is not described as meaningful.

A4. Complete cognitive battery (reviewer 2, comment 3). Restrict the locked rows to participants with all four component scores, keep the primary label, and re-estimate delta-AUC(M3 - M0) and delta-AUC(M4 - M0+) with logistic regression.

A5. Demographically adjusted outcome (reviewer 2, comment 2). Regression-based norms derived within the 3,124 labelled participants: ordinary least squares of the composite z-score on age, age squared, sex, education (categorical) and race and Hispanic origin (categorical). The label is the lowest quartile of the residual, read as performance lower than expected for demographic background. Like the primary label, it is defined on the pooled labelled sample, not per fold. Re-estimate delta-AUC(M3 - M0) and delta-AUC(M4 - M0+) on the locked rows with logistic regression. Externally derived norms are not applied, and this is stated as a limitation.

A6. Outcome-construction descriptives (reviewer 2, comment 3 and Table 1). Missingness by cognitive test with the NHANES reason codes (`CFDCRNC`, `CFDARNC`, `CFDDRNC`, `CFASTAT`), by demographic group; three-score against four-score participants; labelled against unlabelled participants; and the cognitive phenotype of the analytic sample (raw scores, composite distribution and cutoff, language of administration, number of component scores).

A7. Interval for the continuous-outcome analysis. Section 7 item 2 reported the change in cross-validated R-squared without an interval. Add a paired bootstrap 95% CI over participants, same 2,000 resamples, same predictions.

Role	Name	Date
Data and modeling lead	Neil Patel	2026-11-04
Clinical and framing lead	Shreyan Kancharla	2026-11-05

Note added after running Amendment 1 (2026-11-04)

Recorded so that every difference between the amendment above, which was committed before any of its analyses ran, and what was actually run is visible.

- **A5 sample.** The norm model was fitted on the 3,120 labelled participants with complete education data, not all 3,124 (4 have missing education). Age entered centred at 70, with its square. Income was not included, so M0 (which contains income) keeps some discrimination on the adjusted outcome.
- **A3 rule.** The manuscript applies the symmetric 0.02 criterion in its stricter form: a difference is not described as meaningful unless its 95% CI lies wholly beyond ± 0.02. This covers intervals that straddle 0.02 as well as those inside it.
- **A4 and A5 pipeline.** Both used the primary tuned logistic pipeline from `03_model.py`, not the fixed-penalty pipeline from `06_sensitivity.py`.
- **Also run, not listed above.** Decision curves for M4 against M0+ and M0, and the primary specification under the phase 6 pipeline as a reference row for the sensitivity table. Both are descriptive.
- **Outcome.** The demographically adjusted analysis gave a different result from the primary one (delta-AUC(M3 - M0) +0.0362, 95% CI 0.0068 to 0.0667). It is reported as exploratory, in the abstract and conclusions, exactly as specified above.